AWS Network Architecture

An Example of Restructuration and Clean-Up

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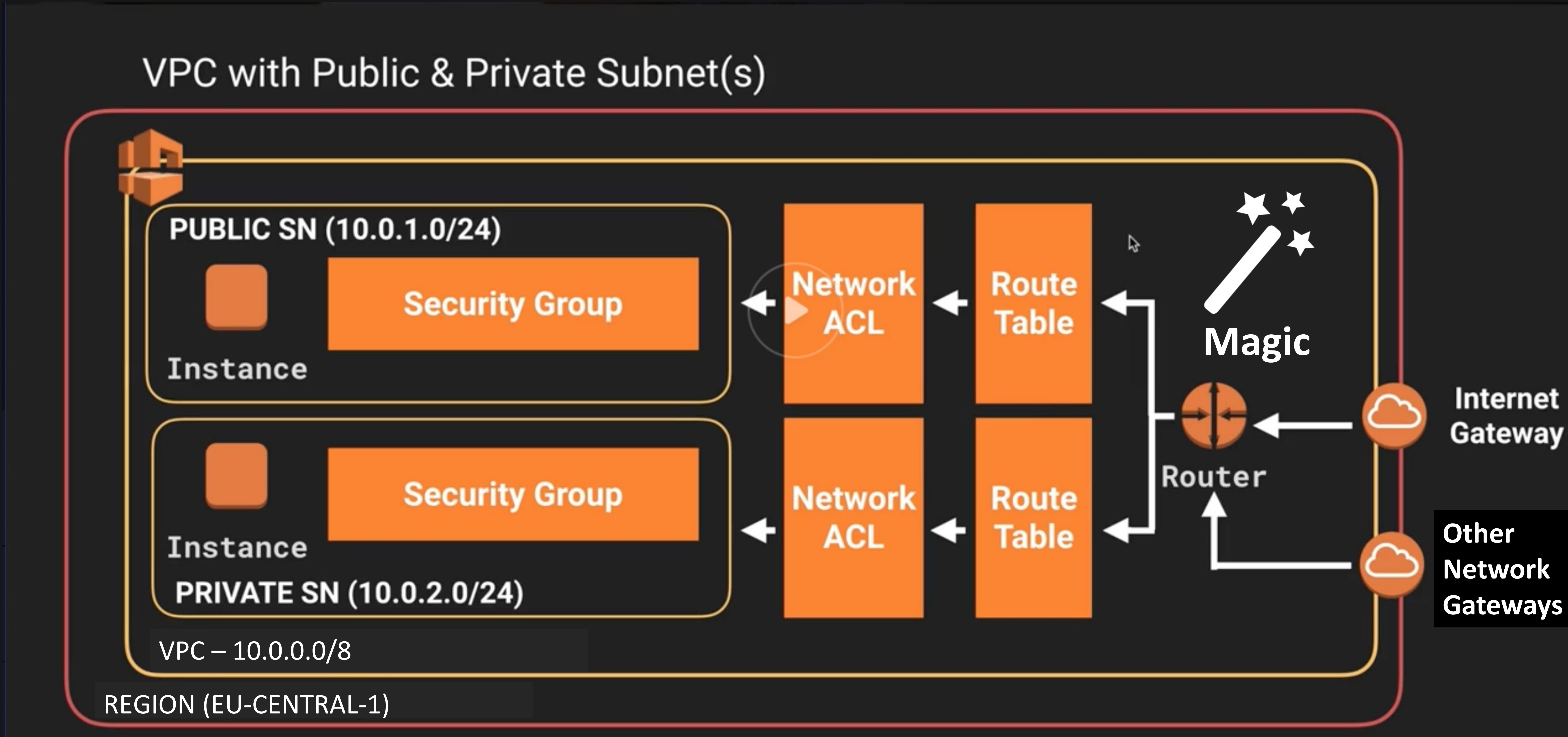
Agenda

- AWS Network
  - Concepts
  - Problems
  - Solutions
  - Restructuration
  - Exception to the Rules
  - Cheat Sheet for Engineers
  - Key Take Away (Reminders)
- Q&A

Intention:

- **Arise curiosity**
- **Give overview**
- **Serve as basic reference for what & where**

AWS Network Basic Concepts



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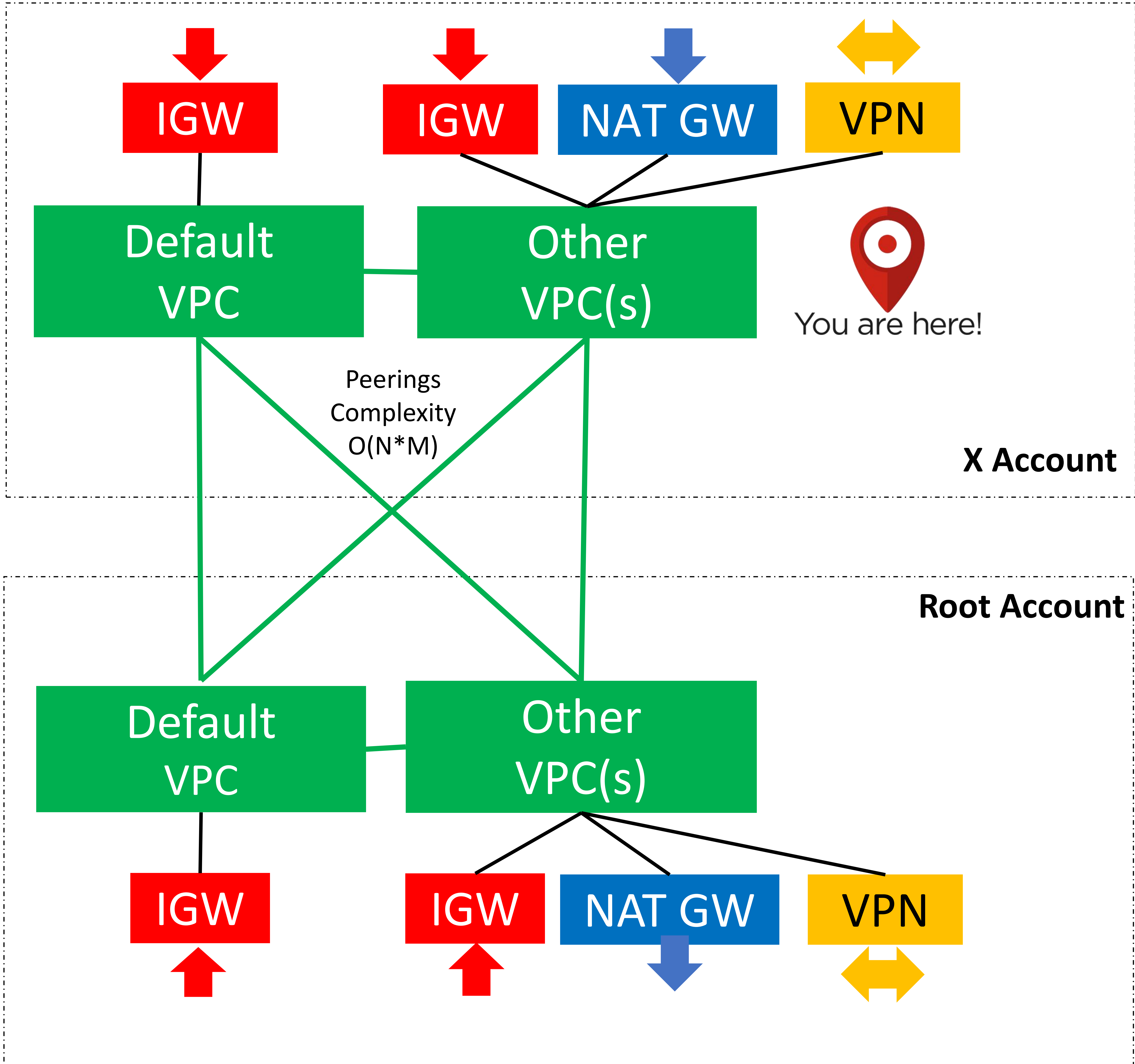
VPC ⇢ Account (Team) ⇢ Organization

Example

Problems:

1. Peering Complexity
2. Multiple Route Tables
3. GW/VPN everywhere
4. Security (Pub./Priv.)
5. Costs (GWs)

Solutions?:



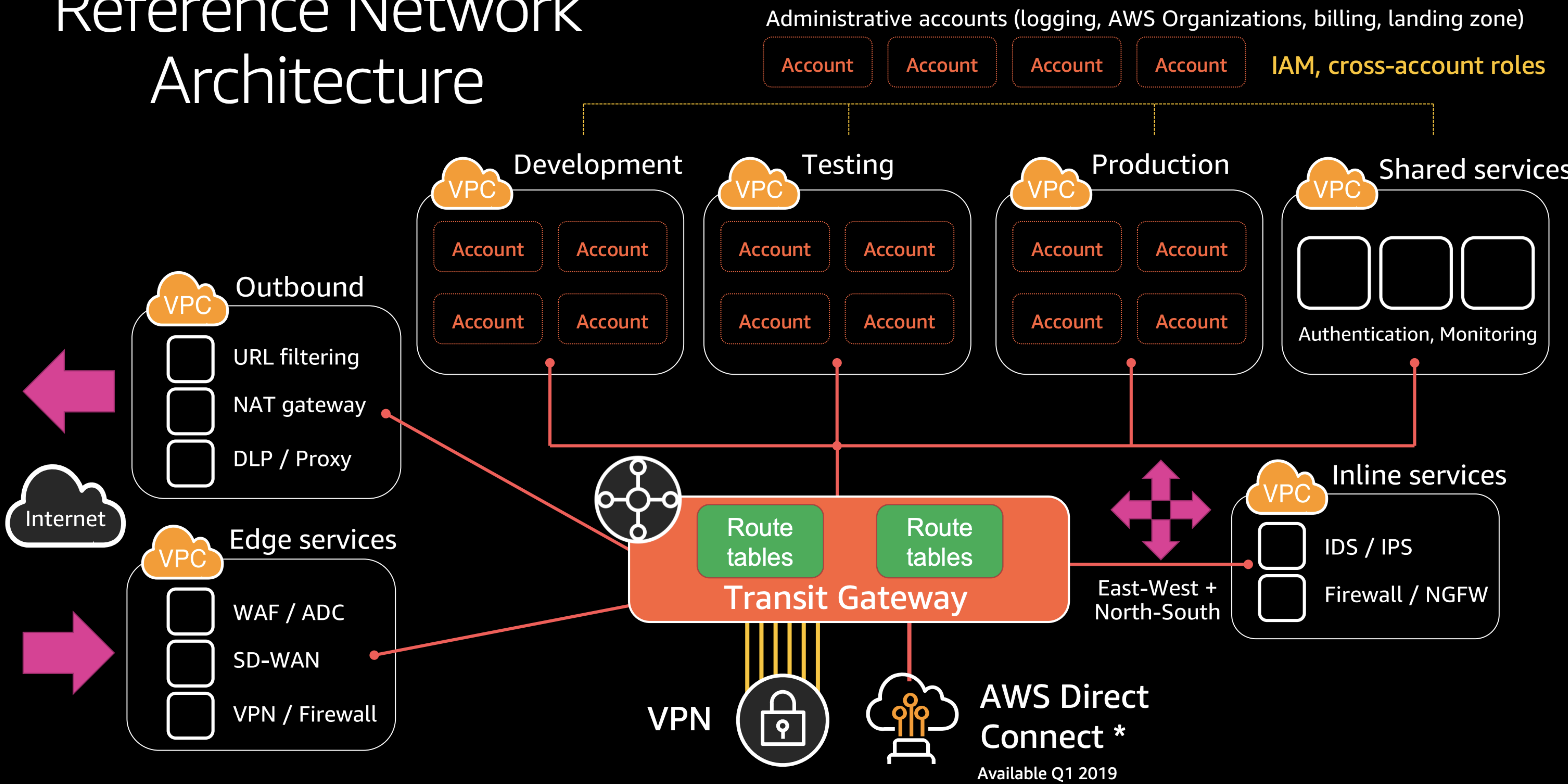
AWS re:Invent

http://aws-reinvent-audio.s3-website.us-east-2.amazonaws.com/2019/2019_presentations.html

1) https://d1.awsstatic.com/events/reinvent/2019/REPEAT_3_From_one_to_many_Diving_deeper_into_evolving_VPC_design_ARC304-R3.pdf

2) https://aws-de-marketing.s3-eu-central-1.amazonaws.com/Field%20Marketing/Summit-Berlin-2019/Presentations/AWS_Summit_Berlin_2019_Feb27_AWS_Networking%20_Advanced_Concepts_%20New_Capabilities.pdf

Reference Network Architecture



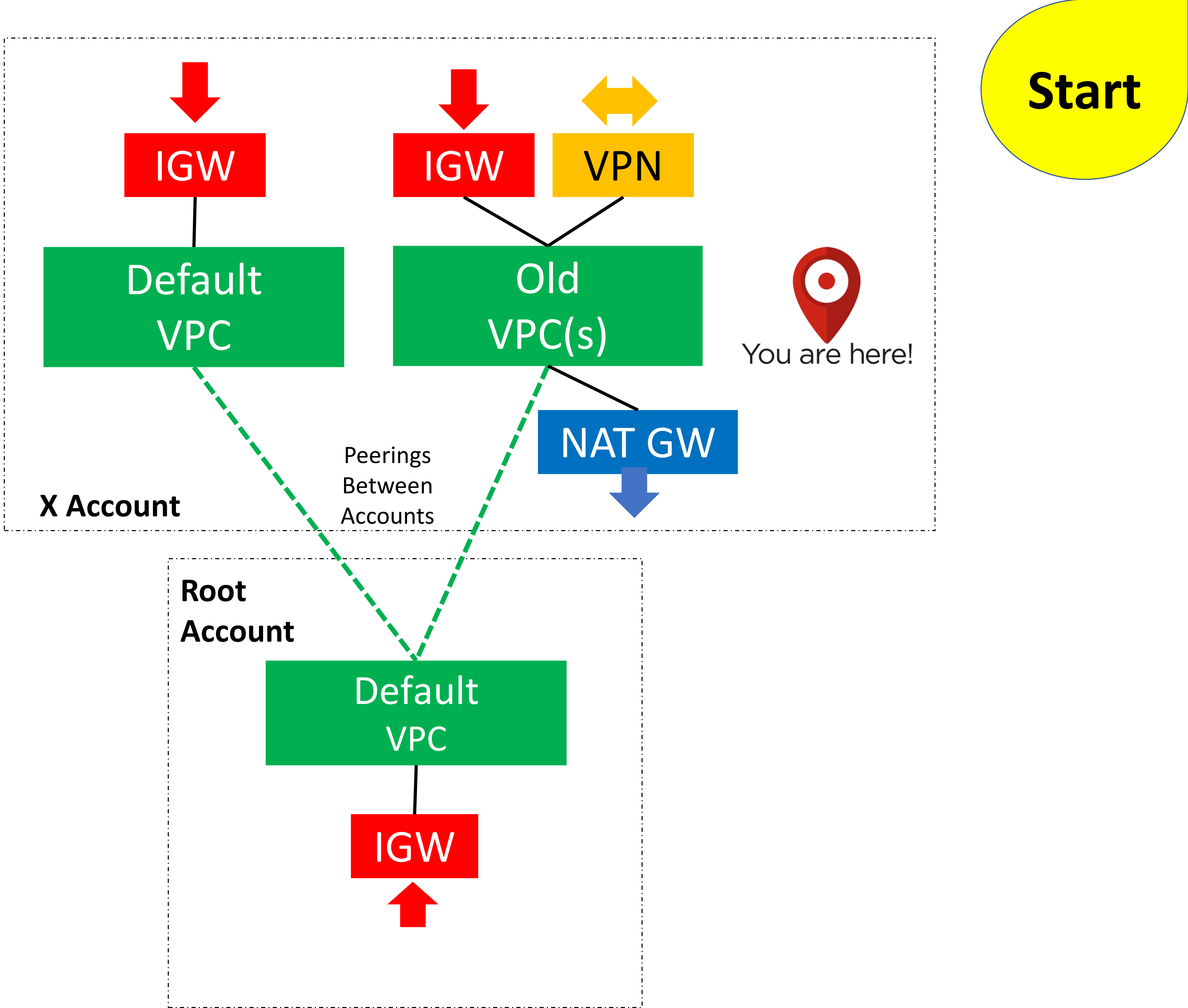
1) https://d1.awsstatic.com/events/reinvent/2019/REPEAT_3_From_one_to_many_Diving_deeper_into_evolving_VPC_design_ARC304-R3.pdf

State-Of-The-Art & Referential => We Took Small Seed!

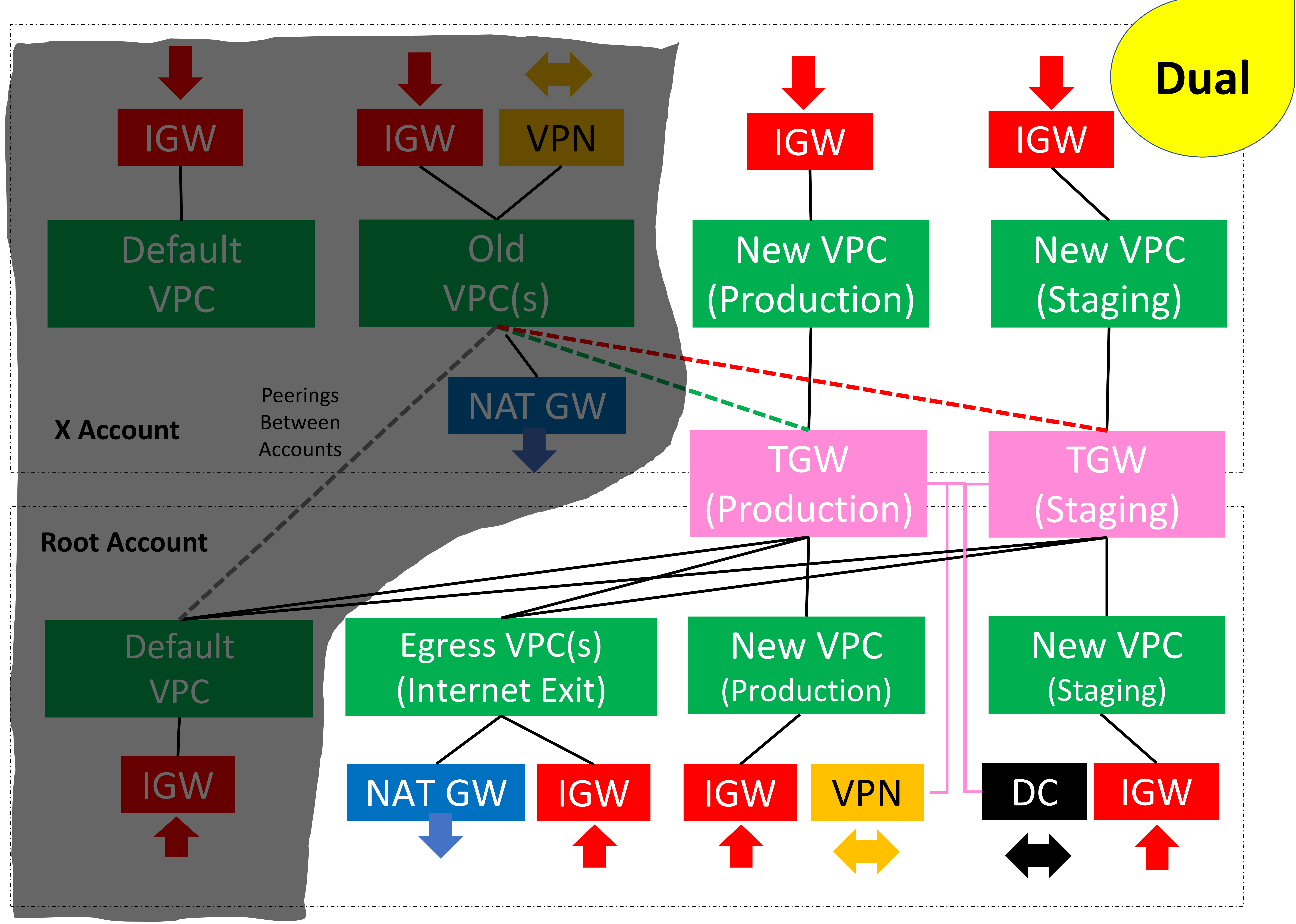
Problems => Solutions

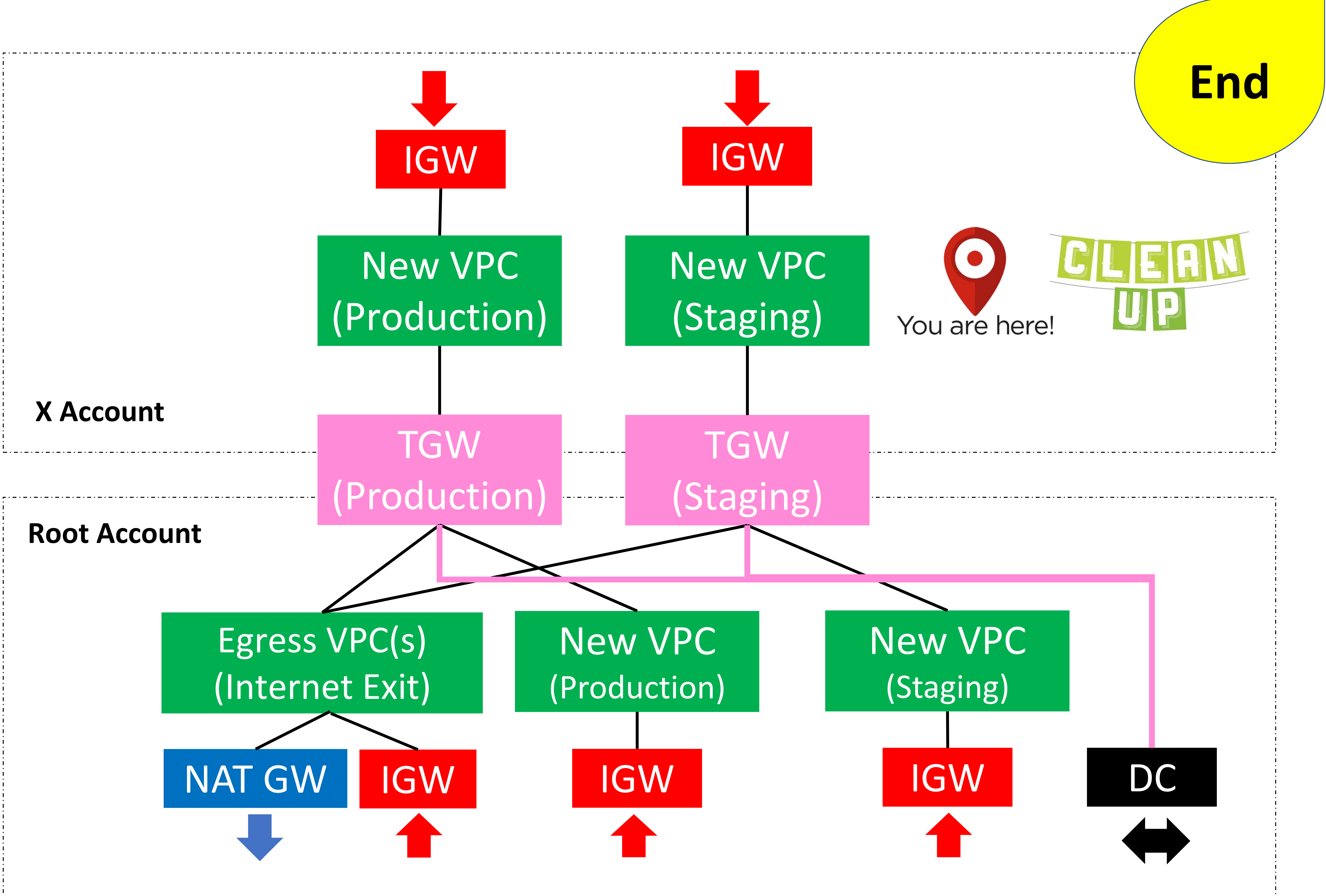
1. **Complexity** => Migrate to TGW
2. **Route Tables** => Central TGW + TF
3. **VPNs** => Carriers + Direct Connect
4. **Security** => Enforce Pub/Priv. Subnets
5. **Costs (GWs)** => Single Internet Exit

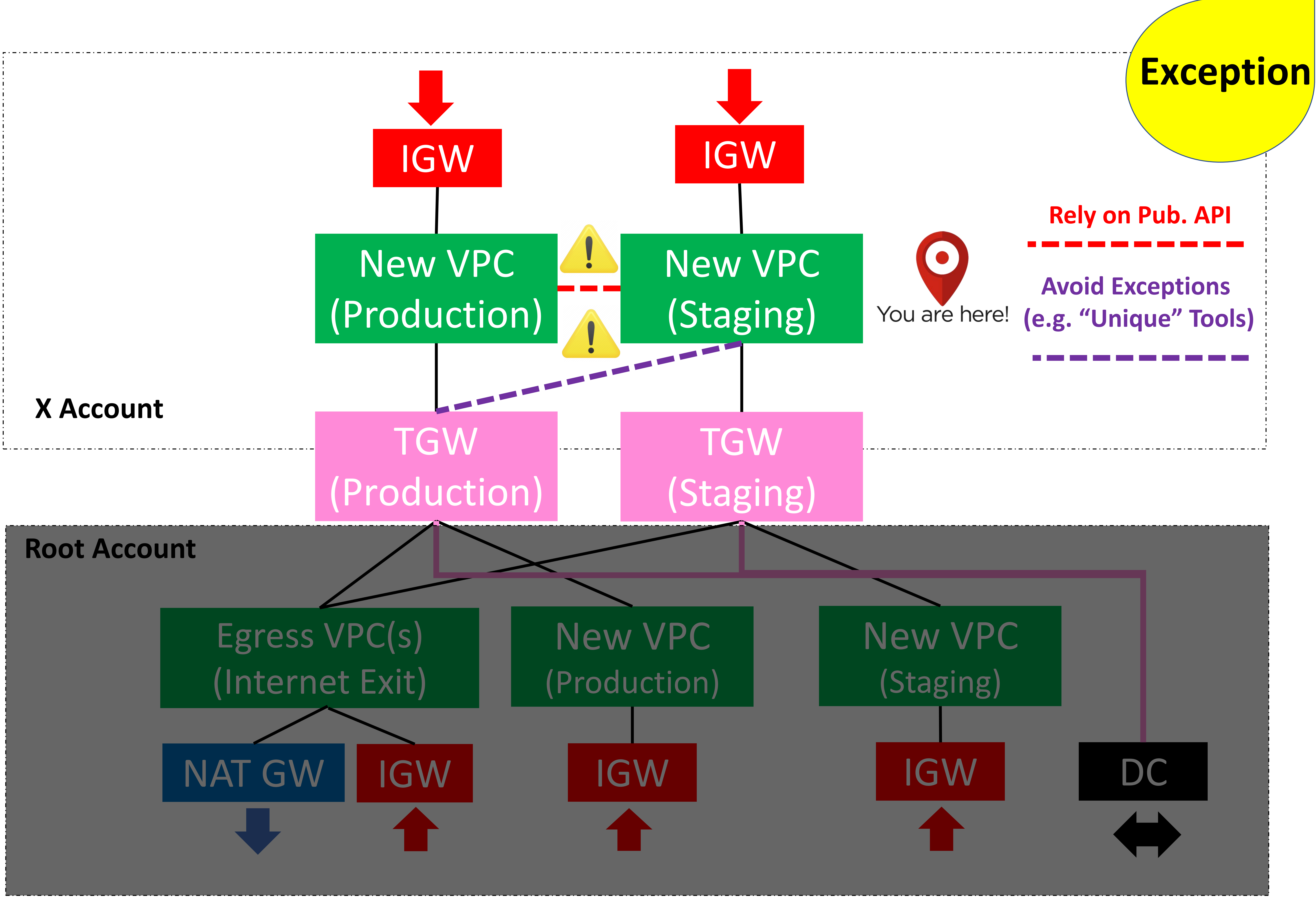
- **ETA/Effort** => O(N), where N ~ Months.
- **Downtime** => Progressive / Phased.
- **Terraform** => Rework / Split.



Start
IGW
IGW
VPN
Default
VPC
Old
VPC(s)
You are here!
NAT GW
Peerings
Between
Accounts
X Account
Root
Account
Default
VPC
IGW

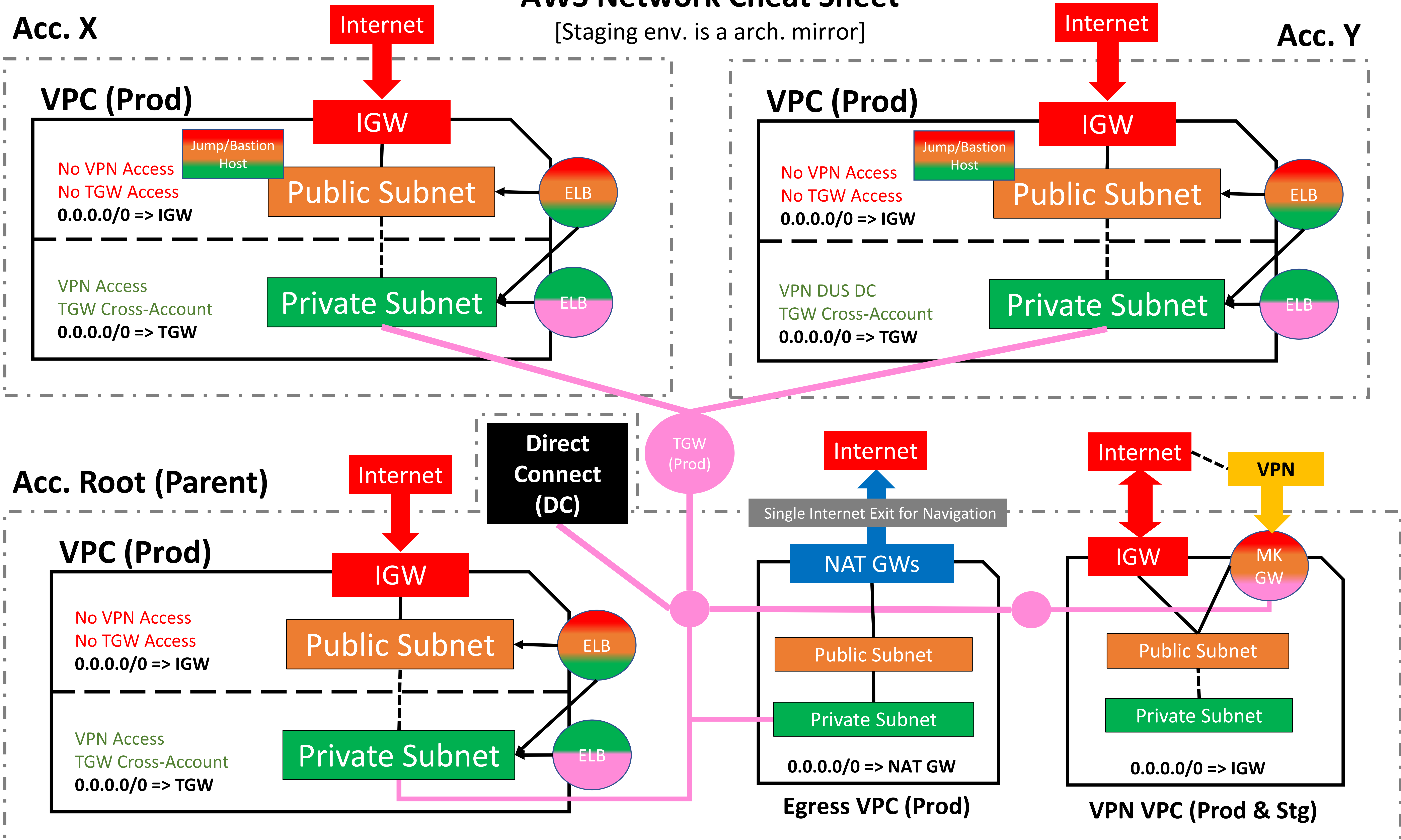

Dual
IGW
IGW
VPN
IGW
IGW
Default VPC
Old VPC(s)
New VPC (Production)
New VPC (Staging)
NAT GW
X Account
Peerings Between Accounts
TGW (Production)
TGW (Staging)
Root Account
Default VPC
Egress VPC(s) (Internet Exit)
New VPC (Production)
New VPC (Staging)
IGW
NAT GW
IGW
IGW
VPN
DC
IGW


End
IGW
IGW
New VPC
(Production)
New VPC
(Staging)
You are here!
CLEAN UP
X Account
TGW
(Production)
TGW
(Staging)
Root Account
Egress VPC(s)
(Internet Exit)
New VPC
(Production)
New VPC
(Staging)
NAT GW
IGW
IGW
IGW
DC


Exception
IGW
IGW
New VPC
(Production)
New VPC
(Staging)
You are here!
Rely on Pub. API
Avoid Exceptions
(e.g. “Unique” Tools)
X Account
TGW
(Production)
TGW
(Staging)
Root Account
Egress VPC(s)
(Internet Exit)
New VPC
(Production)
New VPC
(Staging)
NAT GW
IGW
IGW
IGW
DC

AWS Network Cheat Sheet

[Staging env. is a arch. mirror]



NOTES: 1) Security Groups for resources on public subnets must be restricted to Office IPs unless accessed by customers or external services. For security reasons, PUBLIC SUBNETS ARE NOT ABLE TO REACH neither NAT GWs (instead use the account Internet GW), Transit GW, private VPNs / Direct Connect or any other Cross-Account VPC/Subnets/Resources both public and private. 2). Indeed, public subnets are restricted within the VPC where they are defined and meant to be used for: AWS EC2 instances like jump/bastion hosts, some frontends setups (able to work within the same VPC), Elastic Load Balancers, jump/bastion hosts. Remember, that NAT GWs IP from root (parent) account should be whitelist if external services maintains a whitelist of AWS NAT GW public IPs. 3) Additionally, Jump/bastion hosts can be easily replaced with AWS System Manager https://aws.amazon.com/blogs/mt/replacing-a-bastion-host-with-amazon-ec2-systems-manager to save money and reducing security risks (not urgent).

Key Takeaways - Reminders

1. Save/Print the Cheat Sheet.
2. Do not use Default VPC (If you still have one)
3. Migrate from Old VPCs and prefer new VPCs overall
4. New VPCs (Stage/Production) and 2-3 Availability Zones
5. Don't deploy servers and/or applications on Public Subnets (Use Network/Application ELB instead).
6. New single Internet exit through the AWS Root Account (Whitelist Root Account NAT Gateways IPv4 to external service providers).
7. Whitelist AWS NAT GWs IPs on your Security Groups.